



Adult Primary Care and Pediatric Case Quiz

A 17-year-old, previously healthy male presents via telephone appointment with two days of 101°F fever, sore throat, and a moderate cough. The sore throat bothers him the most, with a pain scale of 7/10.

He is tolerating fluids and able to speak in complete sentences without difficulty.

The parent asks if testing for strep pharyngitis is indicated, or whether you can prescribe antibiotics over the phone.

What is the next best step?

- a. Prescribe antibiotics, given that a sore throat is his most prominent symptom
- b. Obtain further history
- c. Immediately arrange for the Nurse Clinic to conduct strep testing
- d. Immediately deny the request for both strep testing and antibiotics

Best answer: b. Obtain further history

Most episodes of acute pharyngitis are caused by self-limited respiratory viruses, but viral pharyngitis symptoms often overlap with Group A Streptococcus infection. Obtain further history to decide if strep testing is clinically indicated.

Use Centor criteria to help with this decision: (Add 1 point for each symptom)

- ◆ Fever ≥ 100.4°F
- ◆ Absence of cough
- ◆ Tonsillar exudates
- ◆ Tender/swollen anterior cervical lymph nodes

If the Centor score is ≥ 3, recommend strep testing.

What Should We Consider When Diagnosing?

For our patient, the history of fever with the presence of cough resulted in a preliminary Centor score of 1. We should ask further questions about tonsillar exudates and tender cervical lymphadenopathy to guide our decision-making. Clinicians may consider strep testing with a lower Centor score based on clinical judgment.

There are clinical features that are not Centor criteria, but may help determine the need for strep testing:

- ◆ Scarletiform rash
- ◆ Absence of multiple signs and symptoms of viral infections (e.g., rhinorrhea, conjunctivitis, cough, hoarseness, stomatitis, discrete ulcerative lesions or vesicles, diarrhea, etc.)
- ◆ Exposure to an individual with Group A Strep

Beware of strep carriers: Between 5 – 21% of children between 3 – 15 years of age are pharyngeal carriers of strep, and reported rates are 4 – 5% in healthy adults.

In order to prevent “false positives” with pharyngeal strep carriers, avoid strep testing in patients with multiple symptoms **strongly** suggestive of viral illness (e.g., rhinorrhea, conjunctivitis, cough, hoarseness, stomatitis, discrete ulcerative lesions or vesicles, diarrhea, etc.).

What’s Wrong with the Other Options?

- a. For both adult and pediatric patients, prescribing antibiotics without testing is not recommended, even if the Centor score is high. Centor criteria are neither sensitive nor specific for the diagnosis of strep pharyngitis; the use of these criteria does **not** replace testing for GAS and should **not** be used to determine the need for antibiotics. This would also adversely affect our HEDIS measures.
- c. Further history should be obtained to decide if strep testing is truly indicated.
- d. Before declining a request for strep testing or empiric antibiotics, we should obtain as much pertinent history as possible. Patients are more likely to trust our recommendations if they feel we have taken a thorough and thoughtful approach to their questions.

If Nurse Clinic Strep Testing is Needed

- ◆ Place orders for strep testing and antibiotics via the “VMC Pharyngitis” SmartSet, then send to Nurse Clinic for strep protocol. Do **not** notify the patient you have pre-ordered antibiotics. The nurse will notify the patient if rapid strep test is positive; we will have antibiotics immediately available in the pharmacy.
- ◆ [RST locations and clinic hours](#)
- ◆ [Pediatric strep testing locations and workflows](#)

References

Chow, A. W. & Doron, S. (2023). *Evaluation of acute pharyngitis in adults*. Up To Date. <https://www.uptodate.com/contents/evaluation-of-acute-pharyngitis-in-adults>

Shulman, S. T., Bisno, A. L., Clegg, H. W., et al. (2012). [Clinical practice guideline for the diagnosis and management of group A streptococcal pharyngitis: 2012 update by the Infectious Diseases Society of America](#). *Clinical Infectious Diseases*, 55(10), 1279-1282.

Wald, E. R. (2023). *Group A streptococcal tonsillopharyngitis in children and adolescents: Clinical features and diagnosis*. Up To Date. <https://www.uptodate.com/contents/group-a-streptococcal-tonsillopharyngitis-in-children-and-adolescents-clinical-features-and-diagnosis>